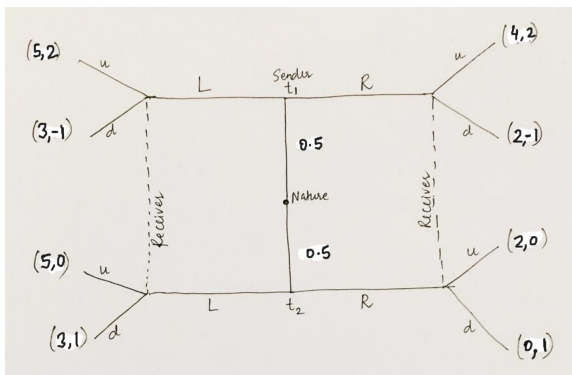
Game Theory (ECO 221): Practice problems

1. Consider a game where nature moves first. Nature selects the type of Sender: type t_1 with probability 0.5 and type t_2 with probability 0.5. Sender knows his type and chooses whether to send a message L or R. Receiver does not know Sender's type, but sees whether Sender played L or R. Receiver then chooses u or d. The game tree for the above game is given by:



The first element of the payoff vector denotes the payoff of Sender while the second element denotes the payoff of Receiver.

(a) Write the pure strategies of the sender. Mention clearly what are his pooling strategies and what are his separating strategies.

(b) Write the pure strategies of the receiver.

(c) Consider Sender's strategy as: Play R when type is t_1 and play R when type is t_2. Can this pooling strategy be sustained as a Perfect Bayesian Equilibrium? Explain the steps properly.

(d) Consider Sender's strategy as: Play L when type is t_1 and play L when type is t_2. Can this pooling strategy be sustained as a Perfect Bayesian Equilibrium? Explain the steps properly.

(e) Consider Sender's strategy as: Play L when type is t_1 and play R when type is t_2. Can this separating strategy be sustained as a Perfect Bayesian Equilibrium? Explain the steps properly.

(f) Consider Sender's strategy as: Play R when type is t_1 and play L when type is t_2. Can this separating strategy be sustained as a Perfect Bayesian Equilibrium? Explain the steps properly.

(g) Consider the semi-separating strategy as: Play L with probability x and play R with probability $1-x$ when the type is t_1 and play L when type is t_2. On the other hand, receiver always choose to play u when the message R is received and mixes by playing u with probability y and d with probability $1-y$ when the message L is received. Can this be sustained as a Perfect Bayesian Equilibrium for any value of x and y? Explain the steps properly.

2. Consider a game where there are two players. Player 2 is one of two possible types: L and H. Type L loves going out with player 1, whereas type H hates it. Player 1 loves going out with player 2 but does not know which type is player 2. She believes that player 2 is type L with probability q and type H with probability $1-q$. The following table gives the payoffs corresponding to each action and type profile, where the first element of any cell represents payoff of Player 1(row player) and second element represents payoff of player 2 (column player).

	B	S
B	(2,1)	(0,0)
S	(0,0)	(1,2)

Player 2 is Type L (with prob q)

	B	S
B	(2,0)	(0,2)
S	(0,1)	(1,0)

Player 2 is Type H (with prob $1-q$)

(a) What are the pure strategies of Player 1?

(b) What are the pure strategies of Player 2?

(c) How many strategy profiles of the form (Player 1's strategy, (Type L's strategy, Type H's strategy)) are there? Write each of these strategy profiles carefully.

(d) Find all Bayesian Nash equilibrium of this game.